Sk. Shafi Masthan Koushik

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EDUCATION

GITAM (Deemed to be University), Hyderabad Campus – Hyderabad, India
Bachelor of Technology, Computer Science and Engineering *Expected 2027*

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Bash, C, C++

AI / ML: PyTorch, ROCm GPU acceleration, ASR pipelines, model inference optimization, Hugging Face

Full-Stack / Web: Node.js, React, REST APIs, Docker, LibreOffice UNO automation

Systems / Linux: Arch Linux, CachyOS, Hyprland, Wayland, GPU driver configuration (AMD RADV/ROCm), shell scripting (fish), performance tuning

Tools: Git, Pi-hole, network configuration, CI-style debugging workflows

PROFESSIONAL EXPERIENCE

AI and Full Stack Development Intern, Swecha Software Foundation (in collaboration with VISWAM.AI and IIIT Hyderabad) – Hyderabad, India June 2026 – July 2026

- Engineered a GPU-accelerated inference pipeline for the Gonthuka ASR volunteer-compute tool on AMD ROCm, resolving Python 3.14 compatibility, HSA_OVERRIDE_GFX_VERSION configuration, and torch symlink integration into a uv-managed virtual environment
- Diagnosed and resolved discrete-GPU wake and permission failures at the sysfs and /opt/rocm ACL level, unblocking confirmed GPU-accelerated inference on 4GB VRAM consumer hardware
- Built a 30-page technical report generation pipeline using the Node.js docx library and LibreOffice UNO automation for programmatic table-of-contents and formatting control
- Contributed full-stack features to Swecha's open-source AI corpus and volunteer-compute platform, tracked across commit history at code.swecha.org/knkssmk
- Worked across AI tooling and full-stack web development workstreams under the supervision of Dr. Jinka Sreedhar, spanning ASR infrastructure and platform features

Technologies: *Node.js, Python, PyTorch, ROCm, Docker, React, LibreOffice UNO, Git*

PROJECTS

Linux Desktop and GPU Performance Engineering

- Rearchitected a dual-GPU (integrated/discrete) rendering pipeline on AMD hardware using DRI_PRIME offload, RADV performance flags, and gamemode integration, eliminating iGPU fallback across a multi-title game library
- Deployed Pi-hole in Docker with custom fish shell profile-switching functions for network-wide ad blocking, toggling automatically between home and campus network configurations
- Migrated a Hyprland/Wayland desktop environment between dotfile frameworks, debugging startup failures traced to missing configuration modules and portal/session-target misconfigurations

Technologies: *Arch Linux, CachyOS, Hyprland, Docker, fish shell, AMD RADV/ROCm*